

HTML DOM Objects with Examples

The HTML DOM (Document Object Model) is a programming interface for HTML and XML documents. It represents the page so that programs can change the document structure, style, and content. The DOM represents the document as nodes and objects. This means that programming languages like JavaScript can interact with the page, modifying elements, attributes, and styles dynamically.

Document

The `document` object is the root of the HTML DOM. It represents the whole web page. Using the `document` object, you can access and manipulate all elements on the page.

Examples:

- `document.getElementById('demo')`: Gets an element by its ID.
- `document.write('Hello')`: Writes directly to the HTML document.
- `document.title`: Gets or sets the title of the document.

Element

An element object represents an HTML element. Once you access an element, you can manipulate it.

Examples:

- `document.getElementById('demo').innerHTML = 'Hi'`: Changes the content.
- `element.setAttribute('class', 'active')`: Sets an attribute.
- `element.classList.add('new')`: Adds a class.

Attributes

Attributes provide additional information about HTML elements.

Examples:

- `element.getAttribute('href')`: Gets the value of an attribute.
- `element.setAttribute('href', 'https://example.com')`: Sets a new value.
- `element.removeAttribute('style')`: Removes an attribute.

Events

DOM events are actions that occur in the browser, such as clicks, mouseovers, or key presses.

Examples:

- `element.addEventListener('click', myFunction)`: Adds a click event listener.
- `element.onclick = function() { alert('Clicked'); }`: Adds inline event.
- `element.removeEventListener('click', myFunction)`: Removes the event listener.

Event Objects

When an event occurs, an event object is created. It contains information about the event.

Examples:

- `event.type`: Returns the type of the event (e.g., 'click').
- `event.target`: Refers to the element that triggered the event.
- `event.preventDefault()`: Prevents default action (like form submission).

HTMLCollection

An HTMLCollection is an array-like object representing a collection of HTML elements.

Examples:

- `document.getElementsByTagName('p')`: Returns a collection of all `<p>` elements.
- `document.forms`: Returns all `<form>` elements.

- Looping: `for (let i = 0; i < collection.length; i++) { console.log(collection[i]); }`

Style

The style object allows you to get or set the CSS styles of HTML elements.

Examples:

- `element.style.color = 'red'`: Sets the text color.
- `element.style.backgroundColor = 'yellow'`: Sets the background color.
- `element.style.display = 'none'`: Hides the element.